



(51) International Patent Classification:

G01N 21/35 (2014.01) G01N 33/24 (2006.01)
A01C 21/00 (2006.01)

(21) International Application Number:

PCT/IB2021/062229

(22) International Filing Date:

23 December 2021 (23.12.2021)

(25) Filing Language:

English

(26) Publication Language:

English

(30) Priority Data:

202011056711 28 December 2020 (28.12.2020) IN

(71) Applicant: **INDIAN INSTITUTE OF TECHNOLOGY KANPUR** [IN/IN]; Dean, Research & Development, Room Number 151, Faculty Building, Post Office: IIT Kanpur, Uttar Pradesh, Kanpur 208016 (IN).

(72) Inventors: **SINGH, Jayant Kumar**; Department of Chemical Engineering, Indian Institute of Technology Kanpur, Uttar Pradesh, Kanpur 208016 (IN). **PRINCE, Pallav**; Department of Chemical Engineering, Indian Institute of Technology Kanpur, Uttar Pradesh, Kanpur 208016 (IN). **AHMAD, Ashar**; Department of Chemical Engineering, Indian Institute of Technology Kanpur, Uttar Pradesh, Kan-

pur 208016 (IN). **KHAN, Mohd Aamir**; Office of Dean of Resources and Alumni, Indian Institute of Technology Kanpur, Uttar Pradesh, Kanpur 208016 (IN). **KHE-MANI, Yashasvi**; Department of Chemical Engineering, Indian Institute of Technology Kanpur, Uttar Pradesh, Kanpur 208016 (IN).

(74) Agent: **MAJUMDAR, Subhatosh** et al.; S. Majumdar & Co., 5 Harish Mukherjee Road, State of West Bengal, Kolkata 700 025 (IN).

(81) Designated States (unless otherwise indicated, for every kind of national protection available): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, IT, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

(84) Designated States (unless otherwise indicated, for every kind of regional protection available): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ,

(54) Title: A PORTABLE SMART DEVICE FOR DETERMINATION OF SOIL NUTRIENTS

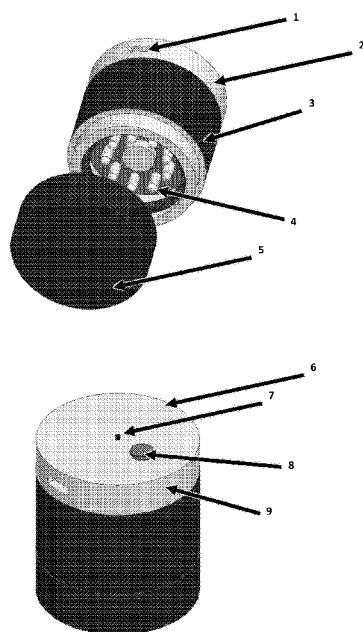


Figure 1

(57) Abstract: Present invention relates to a rapid soil testing device designed for measuring the soil physical and chemical properties. The device is designed to determine and monitor real time soil nutrient content. The device is based on the near infrared (NIR) spectroscopy. Near-IR (NIR) is a spectroscopic strategy which is based on molecular overtones and combination vibrations that originates from the fundamental vibrational bands which are generally found in the mid-IR (MIR) region. The results are displayed over an android smartphone using a developed mobile application and the results can be shared via WhatsApp or email in the form of a summary table. It also has the facility to keep the record of previous recommendations. It is a very low power device having battery backup up-to 60 min in a single charge and can also be powered using the 5 Volt DC supply or the smartphone through OTG cable.



TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

Published:

— *with international search report (Art. 21(3))*

A PORTABLE SMART DEVICE FOR DETERMINATION OF SOIL
NUTRIENTS

TECHNICAL FIELD OF THE INVENTION

5

The present invention relates to a rapid soil testing smart device designed for measuring the physical and chemical properties of soil.

BACKGROUND OF THE INVENTION

10

Agriculture has a vital role in economic growth of India. India's 18% of the GDP comes from the agriculture sector. More than 50% of the total population of India directly depends on the agriculture sector. Soil nutrients are very essential elements for the growth and yield of crops. Soil nutrients are the backbone of the production of the crops and it determines the soil fertility. Sustainable agriculture requires continuous monitoring of the soil health and thus the testing of soil health is very important. India needs innovation in agriculture like the kind of fertilizer to be used and which crop/seed is suitable for the soil so that productivity can be boosted. Excessive use of fertilizer is dangerous not only to the environment but also to the crops as it might cause fertilizer burn. It is estimated that approximately 24 billion tons of fertile soil is lost from agricultural systems every year due to erosion which is a result of poor soil management. And other problems like water pollution, global warming (N₂O contribution), a decrease in microbe's population, high input cost in agriculture are directly linked to it. A routine soil analysis can reduce these negative effects. Therefore, proper soil management and the application of the right fertilizers is essential. Farmers can minimize unnecessary fertilizer expenditure by knowing the exact type and quantity of fertilizers needed for the growth of the crop. Potassium and phosphorus fertilizers are available in a finite amount so these must be used very carefully.

30

The current facilities available in the soil testing laboratories in India require significant time and highly trained workforce and due to these constraints results of the soil analysis get delayed.

- 5 US20160299061A1 relates to Spectrometry systems, methods, and applications.

CN205157419U discloses a soil nutrients detection device based on it is thus clear that - near infrared spectroscopy is technological. Including installing the measurement case room on moving platform, install at the GPS module and the
10 liquid crystal display module at moving platform top and install generator and the three -point linkage at the moving platform front end, measure case five face seals of room, bottom opening and soil surface contact, light source are installed and are being measured the indoor top of case, and the fiber optic probe is installed and is being measured the indoor centre of case, the fiber optic probe is through
15 spectrum appearance connection director, and GPS module and liquid crystal display module all are connected to the controller, and three -point linkage is connected to the tractor. The utility model discloses a soil nutrients detects in succession, and configuration operation is simple and convenient, and low cost easily promotes, and whole testing process is pollution -free, is applicable to carry
20 out incessant detection in succession in the large tracts of land farmland. However, this device requires to be installed on a mobile platform. Further, the testing of dry soil is not facilitated by this device.

US8767194B2 discloses a soil analysis device and a method for measuring
25 characteristics of a soil sample. A mixing chamber combines a soil sample and an extractant into a liquid mixture. A filtration system receives and filters the liquid mixture. The filtered liquid mixture is transmitted to a measurement cell. The measurement cell is coupled to a light source so that light propagating from the light source is attenuated and/or reflected by the liquid mixture and is measured
30 by one or more optical detectors that are also coupled to the measurement cell. The optical detector generates an attenuation spectrum and/or a reflection

spectrum indicating light received by the one or more detectors at different wavelengths. The attenuation and/or reflection spectrum is used to determine the characteristics of the soil sample.

- 5 CN Patent No. 103091502A disclose a mobile device for measuring soil fertility which utilises a multichannel ion selective electrode method.

CN Patent No. 102435656A relates to quick detection of soil nutrient and discloses a sensor array which comprises a silver-silver chloride reference
10 electrode and multiple polypyrene polymer sensitive film sensors for detecting soil nitrate nitrogen.

US Patent No. 5887491A relates to a soil analysis system for determining various soil characteristics which may include moisture content, organic matter content
15 and the presence of nitrogen phosphate, potassium and other elements. Different testing assemblies includes: reflectance testing, electrophoresis testing and chromatography testing.

US10109024B2 relates to Collecting data to generate an agricultural prescription.
20

CN Patent No. 1427262A provides automatic soil sampling and analysis method using a robotic vehicle comprising of a ground drive system of the robot in the field. Soil samples are transported to a laboratory on the micro-robots for analysing the soil.

25

CN Patent No. 206710364U provides a pair of portable readily available nutrient of soil detection device, it includes a glass microelectrode array and utilizes modified electrode detection techniques.

30 US10031076B2 discloses mobile smart device infrared light measuring apparatus, method, and system for analysing substances. The device is a hand-held

spectroscopy device that may be integral to a mobile phone or smart device such as a smart phone, tablet, personal digital assistant, computer or other device that is portable and capable of performing applications. A liquid sample port internal to the device and in close proximity to the device spectrometer performs infrared spectra analysis on liquid samples, allowing both portability as well as highly sophisticated and specific spectral analysis of liquid samples. The device has wireless communication capability, enabling transmission of data and spectral imagery across the globe.

10 CA2822876C related to the analyte ion detection method and device.

US20160216245A1 discloses Infrared aerial thermography for use in monitoring plant health and growth.

15 JP2017505901A related to the spectroscopic system, method and application.

However, there remain a need for a device which can address the issue of quantifying the required amount of fertilizers. The current facilities available in the soil testing laboratories in India require significant time and highly trained workforce and due to these constraints results of the soil analysis get delayed. A smart device is needed to resolve all the above mentioned issues.

SUMMARY OF THE INVENTION

25 The following disclosure presents a simplified summary of the invention in order to provide a basic understanding of some aspects of the invention. This summary is not an extensive overview of the present invention. It is not intended to identify the key/critical elements of the invention or to delineate the scope of the invention. Its sole purpose is to present some concept of the invention in a simplified form as a prelude to a more detailed description of the invention presented later.

The object of present invention is to overcome the problems of prior arts.

Another object of present invention is to develop a rapid soil testing device
5 designed for measuring the soil physical and chemical properties.

The device is designed to determine and monitor real time soil nutrient content. The device is based on the near infrared (NIR) spectroscopy. Near-IR (NIR) is a spectroscopic strategy which is based on molecular overtones and combination
10 vibrations that originates from the fundamental vibrational bands which are generally found in the mid-IR (MIR) region. The results are displayed over an android smartphone using a developed mobile application and the results can be shared via WhatsApp or email in the form of a summary table. It also has the facility to keep the record of previous recommendations. It is a very low power
15 device having battery backup up-to 60 min in a single charge and can also be powered using the 5 Volt DC supply or the smartphone through OTG cable.

One aspect of present invention relates to a portable smart device for measuring the physical and chemical properties of soil. The device comprises (1) an upper
20 part to cover a main body from the top, (ii) a middle part and (iii) a lower part. The upper part further comprises a Header Cap with Power Button, a Reset Button, a Sensor Pin and a USB port assembled. The middle part further comprises the Main Body with optical and control part assembled and the lower part further comprises a sample and calibration cup. Slots for USB ports and a pin
25 port are provided at the top at the upper part. The upper part is mechanically connected with the main body by using plurality of screws. The middle part further comprises a control unit (battery, microcontrollers etc.) and optical unit (LEDs, Photodiodes).

30 The device further comprises a bluetooth module circuit, battery circuit and other sensors circuit .

Optical circuit contains LEDs which acts as optical source and a photodiode which acts as receiver of the reflected light.

5 A buzzer is also added to the circuit to indicate successful completion of different processes.

In the mobile application, an option to store the fertilizers and crop data is incorporated so that the optimized amount of fertilizer needed for a particular crop with specified field size can be recommended. This device has the potential to
10 replace the traditional wet labs and current analytical methods used in soil testing laboratories in India for measuring the nutrient amount and fertilizer recommendation as it is easy to use, portable, cost effective, consumes less time, accurate and reliable. Farmers do not require any sophisticated training to operate the device. As it is compatible with android devices it can be modified to add
15 geographical position (GPS) and hence, a nationwide soil profile mapping can be obtained by uploading the soil health data which can help in policy formulation for fertilizer usage.

Other aspects, advantages, and salient features of the invention will become
20 apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses exemplary embodiments of the invention.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

25

The above and other aspects, features and advantages of the embodiments of the present disclosure will be more apparent in the following description taken in conjunction with the accompanying drawings, in which:

30 Figure 1 illustrates overall design of the device (a) 3D Cross View (b) 3D Top View as disclosed in one of the embodiment of present invention.

Figure 2 illustrates design of upper part (a) Top View (b) Dimensions as disclosed in one of the embodiment of present invention.

- 5 Figure 3 illustrates design of middle part as disclosed in one of the embodiment of present invention.

Figure 4 illustrates optical unit of the device (LEDs assembly) as disclosed in one of the embodiment of present invention.

10

Figure 5 illustrates design of lower part as disclosed in one of the embodiment of present invention.

- 15 Figure 6 illustrates block diagram of electrical circuit as disclosed in one of the embodiment of present invention.

Figure 7 illustrates electrical circuit diagram as disclosed in one of the embodiment of present invention.

- 20 Figure 8 illustrates working device with the mobile application.

- Graph 1 illustrates (a) Relationship between the laboratory analysis (actual values) and prediction values of train set for soil nitrogen (b) Relationship between the laboratory analysis (actual values) and prediction values of test set for soil
25 nitrogen.

- Graph 2 illustrates (a) Relationship between the laboratory analysis (actual values) and prediction values of train set for soil phosphorus (b) Relationship between the laboratory analysis (actual values) and prediction values of test set for soil
30 phosphorus.

Graph 3 illustrates (a) Relationship between the laboratory analysis (actual values) and prediction values of train set for soil organic carbon (OC) (b) Relationship between the laboratory analysis (actual values) and prediction values of test set for soil organic carbon (OC).

5

Graph 4 illustrates (a) Relationship between the laboratory analysis (actual values) and prediction values of train set for clay content (b) Relationship between the laboratory analysis (actual values) and prediction values of test set for clay content.

10

Graph 5 illustrates (a) Relationship between the laboratory analysis (actual values) and prediction values of train set for CEC (b) Relationship between the laboratory analysis (actual values) and prediction values of test set for CEC.

15 Graph 6 illustrates (a) Relationship between the laboratory analysis (actual values) and prediction values of train set for soil potassium (b) Relationship between the laboratory analysis (actual values) and prediction values of test set for soil potassium.

20 Graph 7 illustrates (a) Stability of optical units for soil samples and (b) Stability of optical units for white sheet (reference).

Persons skilled in the art will appreciate that elements in the figures are illustrated for simplicity and clarity and may not have been drawn to scale. For example, the dimensions of some of the elements in the figure may be exaggerated relative to other elements to help to improve understanding of various exemplary embodiments of the present disclosure. Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.

30

DETAILED DESCRIPTION OF THE PRESENT INVENTION

The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the present disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding, but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the present disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the present disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the present disclosure is provided for illustration purpose only and not for the purpose of limiting the present disclosure as defined by the appended claims and their equivalents.

It is to be understood that the singular forms "a," "an," and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, a reference to "a component surface" includes a reference to one or more of such surfaces.

All terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which various embodiments belong. Further, the meaning of terms or words used in the specification and the claims should not be limited to the literal or commonly employed sense but should be construed in accordance with the spirit of the disclosure to most properly describe the present disclosure.

The terminology used herein is for the purpose of describing particular various embodiments only and is not intended to be limiting of various embodiments. As used herein, the singular forms "a," "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising" used herein specify the presence of stated features, integers, steps, operations, members, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, members, components, and/or groups thereof. Also, expressions such as "at least one of," when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

The present disclosure will now be described more fully with reference to the accompanying drawings, in which various embodiments of the present disclosure are shown. The present disclosure may, however, be embodied in many different forms and should not be construed as being limited to the various embodiments set forth herein, rather, these various embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the present disclosure. Furthermore, a detailed description of other parts will not be provided not to make the present disclosure unclear. Like reference numerals in the drawings refer to like elements throughout.

Present invention relates to a rapid soil testing device designed for measuring the soil physical and chemical properties. The device is designed to determine and monitor real time soil nutrient content. The device is based on the near infrared (NIR) spectroscopy. Near-IR (NIR) is a spectroscopic strategy which is based on molecular overtones and combination vibrations that originates from the fundamental vibrational bands which are generally found in the mid-IR (MIR) region. The results are displayed over an android smartphone using a developed mobile application and the results can be shared via WhatsApp or email in the

form of a summary table. It also has the facility to keep the record of previous recommendations. It is a very low power device having battery backup up-to 60 min in a single charge and can also be powered using the 5 Volt DC supply or the smartphone through OTG cable.

5

One aspect of present invention relates to a portable smart device for measuring the physical and chemical properties of soil. The device comprises (1) an upper part to cover a main body from the top, (ii) a middle part and (iii) a lower part. The upper part further comprises a Header Cap with Power Button, a Reset
10 Button, a Sensor Pin and a USB port assembled. The middle part further comprises the Main Body with optical and control part assembled and the lower part further comprises a sample and calibration cup. Slots for USB ports and a pin port are provided at the top at the upper part. The upper part is mechanically connected with the main body by using plurality of screws. The middle part
15 further comprises a control unit (battery, microcontrollers etc.) and optical unit (LEDs, Photodiodes).

The device further comprises a bluetooth module circuit, battery circuit and other sensors circuit .

20

Optical circuit contains plurality of LEDs which acts as optical source and a photodiode which acts as receiver of the reflected light.

A buzzer is also added to the circuit to indicate successful completion of different
25 processes.

The device (refer Figure 1) can be divided into three parts such as:

1. Upper Part: Header Cap (2) with USB port (1), Sensor Pin (6), Reset Button (7), and Power Button (8) assembled
- 30 2. Middle Part: Main Body (3) with optical part (4) and control part (9) assembled

3. Lower Part: Calibration cum Sample Cup (5)

Component 1: Upper Part (Figure 2)

The upper portion of the device consists of USB ports (1), sensors port (6), Reset Button (7), Power Button (8), and battery charging indicator. The main purpose of the upper part is to cover the main body from the top. The diameter of the upper cap is 67 mm with a thickness of 20 mm. Slots for USB ports and a pin port are provided at the top. It is connected with the main body using two screws of diameter 3mm.

10 **Component 2: Middle Part (Figure 3)**

The middle part of the device consists of a control unit (battery, microcontrollers etc.) and optical unit Infrared LEDs, InGaAs Photodiode (refer Figure 4). The shape of the main body is hollow cylindrical with outer and inner diameter of 67 mm and 50 mm respectively. A quartz glass of diameter 52 mm and thickness 1.5 mm is also attached at the bottom of the mid part which prevents sand particles from entering the main circuit and protects the optical unit of the device.

Component 3: Lower Part (Figure 5)

Calibration cum sample cup are the bottom most part of the device. This is the part where soil samples are placed and scanned. During Calibration, instead of soil sample white sheet is placed and scanned. White sheet is used as a reference for the LEDs. The inner and outer diameters of the sample cup are 67 mm and 62 mm respectively.

25 Referring Figure 6, when switch is ON first of all power 5V go into Arduino Nano board and Bluetooth module (HC 05) the first LED will turn ON then reflected light will be detected by InGaAs sensor and corresponding voltage will be stored by Arduino Nano and one by one each LEDs will turn ON and corresponding InGaAs sensor voltage reading will read by Arduino Nano board.

The electrical circuit, referring Figure 7, of the device includes Optical circuit (LEDs assembly and Photodiode), Control circuit (Microcontroller circuit), Bluetooth module circuit, other sensors circuit and battery circuit. The main function of these circuits is to record the voltage and transmit it to the smartphone via Bluetooth which is further used for soil nutrient estimation. Optical circuit contains LEDs which acts as optical source and a photodiode which acts as receiver of the reflected light. Various logical and arithmetic operations are required to be performed on the recorded voltage. So, a microcontroller is required to perform these functions. A buzzer is also added to the circuit to indicate successful completion of different processes.

The main function of the device is to record the voltage and transmit it to the smartphone via Bluetooth which is further used for soil nutrient estimation. The most important part of the device is Optical Unit includes nine different wavelength infrared LEDs and one Near-infrared InGaAs photodiode. All these components were assembled on a single round PCB having a diameter of 51 mm and thickness of around 2 mm.

The microcontroller (AT mega 328) was used to control the sensor which was installed in it and IEEE Std 802.15.1 used for data communication between device and smartphone application. Total of nine Infrared LEDs and one NIR sensor was used to develop this device. First white sheet was used to calibrate the device. Each LED was switched on for 10-12 seconds and the corresponding reflected IR light received by NIR sensor also known as photodiode sensor and reflectance voltage reading gets stored in the mobile application

Results:

Various traditional methods such as the Kjeldahl method, and the flame photometer etc. are employed in the determination of various soil properties. These traditional laboratory methods are too costly and time consuming and often impractical in many regions. Due to the storage and uneven distribution of Soil

Testing Labs (STLs) across the nation farmers have to wait for longer to receive their soil health report and which further leads to delay in purchasing of crops and fertilizers. These challenges led to the need of rapid soil testing methods and devices.

5

For the purpose of the device development the reference soil nutrients were determined using these traditional laboratory methods. The nitrogen was determined using the Kjeldahl setup, the organic carbon using the Wakley-Black method, the phosphorus using the Olsen method, potassium was determined using the flame photometer apparatus, clay content was determined using the hydrometer method and CEC was determined using the Ammonium acetate (NH₄OAc) method. Predictive models were developed using these reference soil nutrients and the absorbance reading obtained from the device. Graphs 1-7 show the model developed and comparison with results of the well-known established methods.

15

Cation Exchange Capacity

An indirect approach called Pedotransfer function (PTF) was developed using Multivariate non-linear regression model to determine the CEC of soil. In Pedotransfer function other spectrally active properties such as Organic Carbon and clay content were used as independent variables for predicting the CEC value. There exists a strong correlation between the CEC, OC and clay which can be seen from Table 1.

20

Table 1: Pearson correlation between CEC, OC, clay, Silt and Sand

Variables	CEC	OC	Clay	Silt	Sand
CEC	1	-	-	-	-
OC	0.698*	1	-	-	-
Clay	0.806*	0.648*	1	-	-

Silt	0.177	0.286	0.131	1	-
Sand	-0.272	-0.359**	-0.252	-0.992*	1

* Correlation is significant at the 0.01 level ($P < 0.01$)

* Correlation is significant at the 0.05 level ($P < 0.05$)

Soil Potassium

- 5 Soil Potassium was estimated using the CEC and clay content of each sample. Graph 6 shows the predicted value from the device and actual potassium from the standard measurement.

Stability and Repeatability of device

- 10 Stability of the optical units which consists of LEDs of different wavelengths were examined. Reading of these LEDs were recorded for soil samples shown in Graph 7 (a) and white sheet (reference) shown in Graph 7(b) on different days to verify the stability of these LEDs. Low fluctuations in voltage were noted in all the LEDs which suggest that the optical unit of the device is stable.

15

NIR Spectroscopy plays a significant role in agriculture and shows a tremendous potential to be a suitable technique for rapid and affordable estimation of various soil parameters. The developed device is based on the same principle and can be used as an alternative for existing traditional laboratory soil testing methods. It can be of great help to the end users like farmers as it provides real time on-field results and accordingly decides the judicious use of fertilizers.

20

The major advantages of the developed device are rapid, portable, handheld, simultaneous determination of multiple parameters, no prior sample preparation and zero chemical consumption. Additional sensors such as moisture sensor and EC (Electroconductivity) and pH meter can also be connected to the device externally to measure parameters. Overall this device will help farmers in boosting their productivity and yield, thus increasing their productivity.

25

CLAIMS:

1. A portable smart device for measuring the physical and chemical properties of soil, comprising (i) an upper part to cover a main body (3) from the top, (ii) a middle part and (iii) a lower part; wherein the upper part comprises a Header Cap (2) with Power Button (8), a Reset Button (7), a Sensor Pin (6) and a USB port (1) assembled, the middle part comprises the Main Body with an electrical circuit assembled and the lower part comprises a sample and calibration cup (5) to contain a soil sample.
2. The portable smart device as claimed in claim 1, wherein the electrical circuit includes Optical circuit (4), Control circuit (9), Bluetooth module circuit, other sensors circuit and battery circuit; wherein the optical unit includes plurality of infrared LEDs as optical source and one Near-infrared InGaAs photodiode sensor as receiver of reflected light in form of reflected voltage when light incidents from the LED lights on the soil sample and reflects from it; and wherein the control circuit includes a microcontroller to perform logical and arithmetic operations on the reflected voltage and transmit the reflected voltage data to a smartphone which is configured to receive the voltage data via Bluetooth of Bluetooth module circuit and further use for soil nutrient estimation.
3. The portable smart device as claimed in claim 1, wherein Slots for USB ports and a pin port are provided at the top at the upper part.
4. The portable smart device as claimed in claim 1, wherein the upper part is mechanically connected with the main body by using plurality of screws.

5. The portable smart device as claimed in claim 1, wherein a buzzer is also added to the circuit to indicate successful completion of different processes.

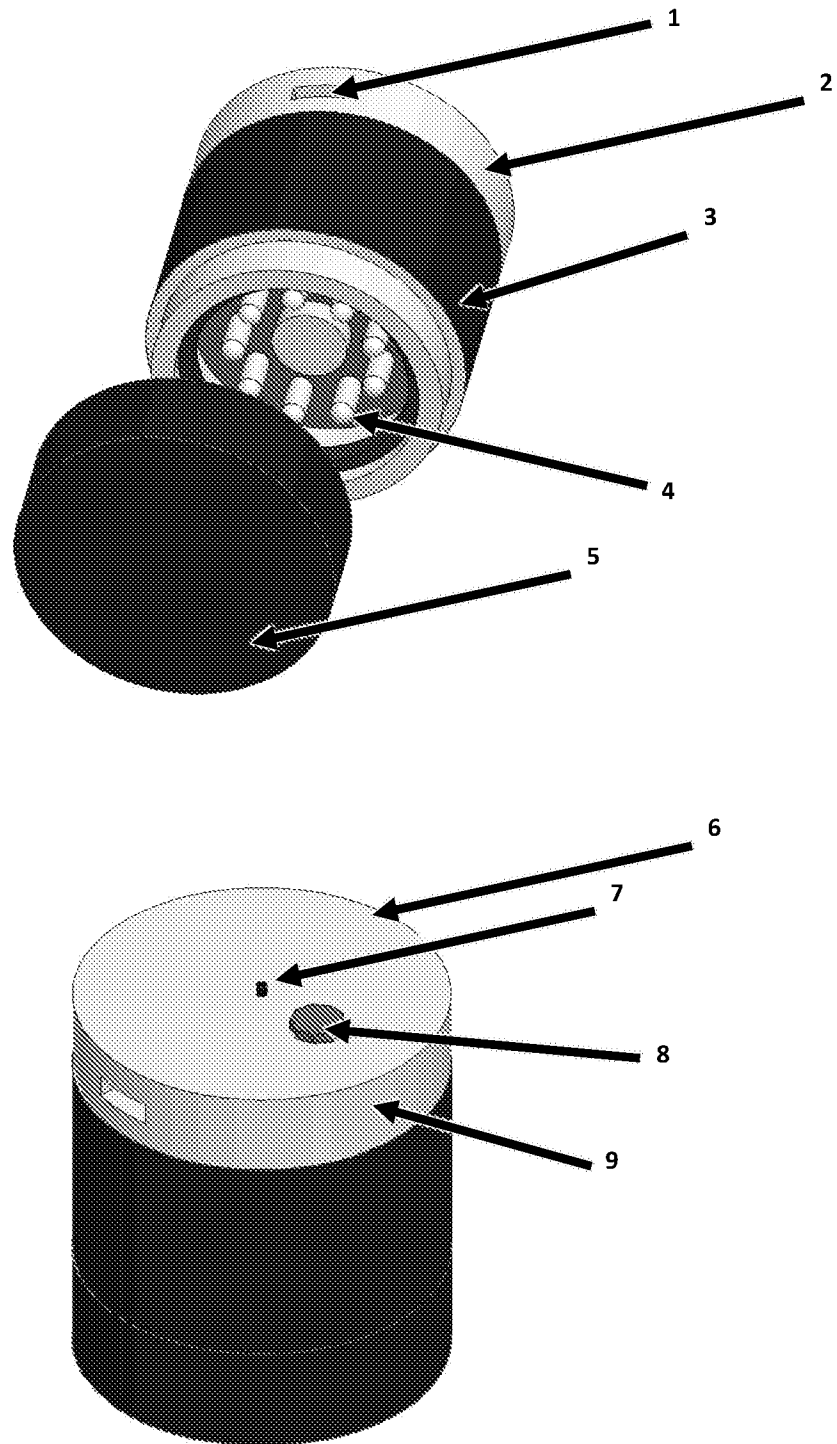


Figure 1

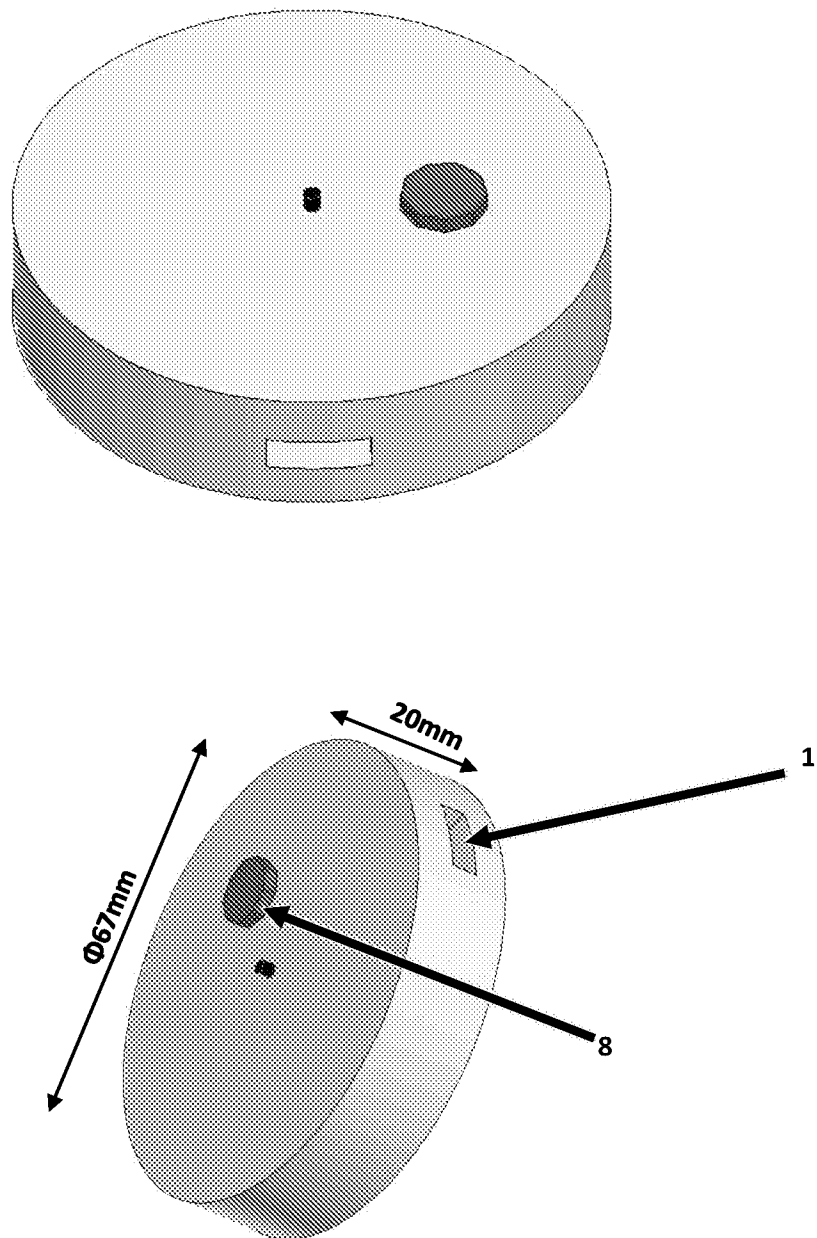


Figure 2

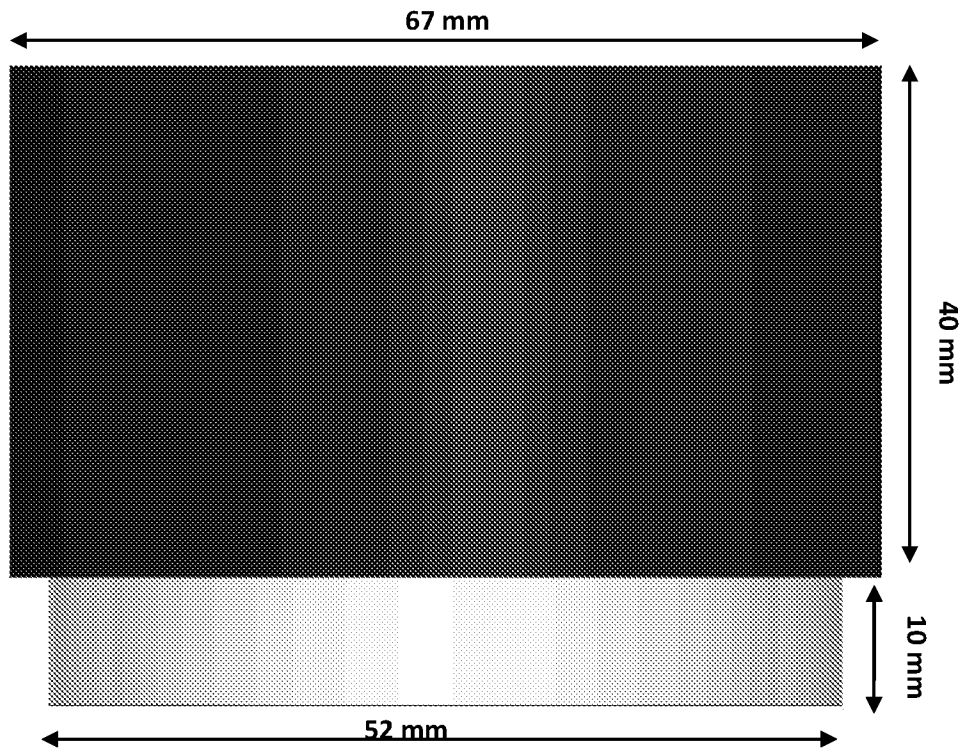


Figure 3

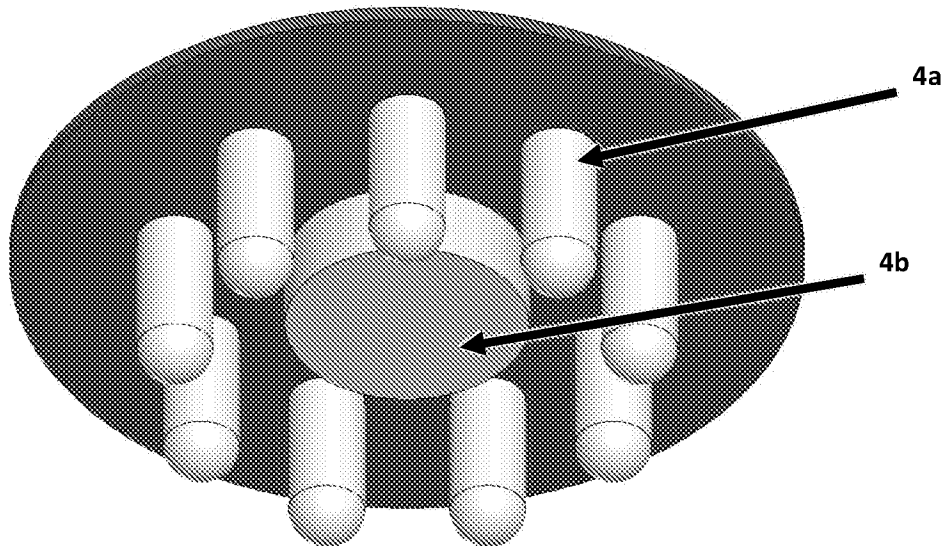


Figure 4

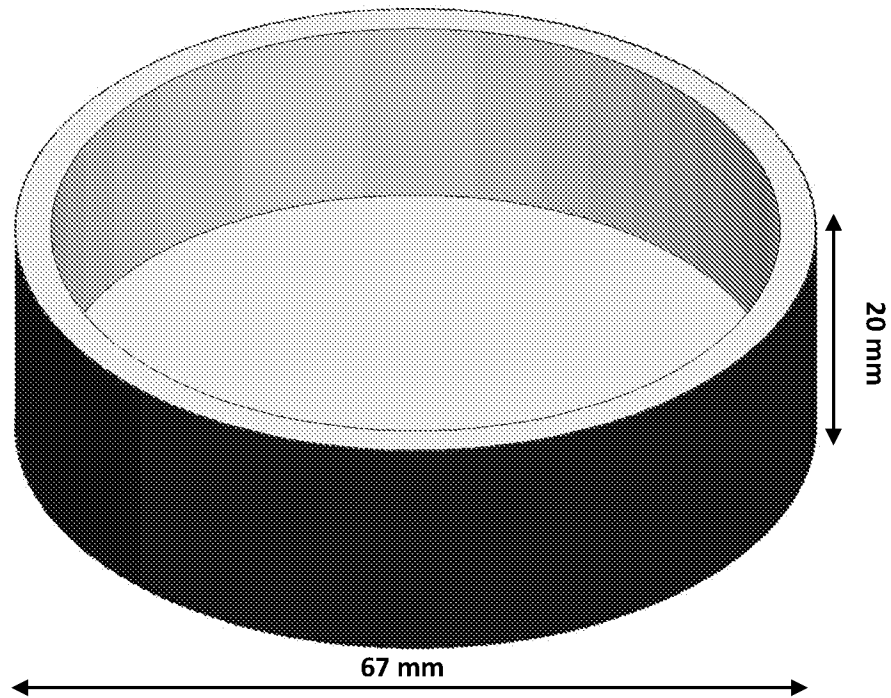


Figure 5

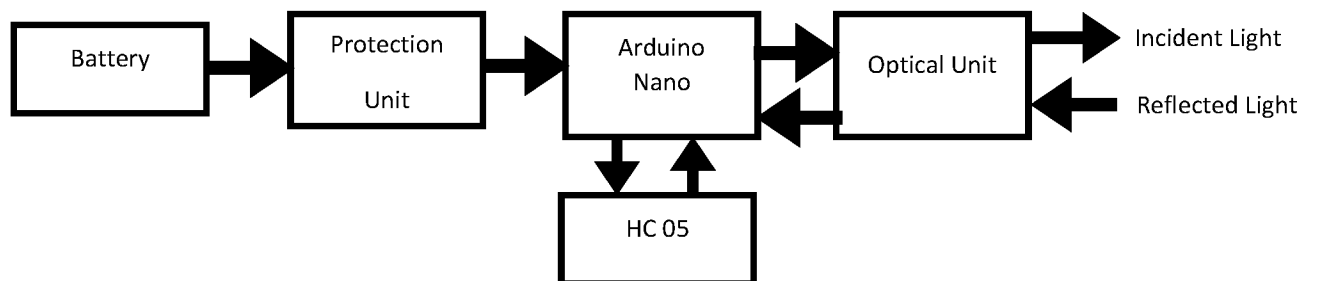


Figure 6

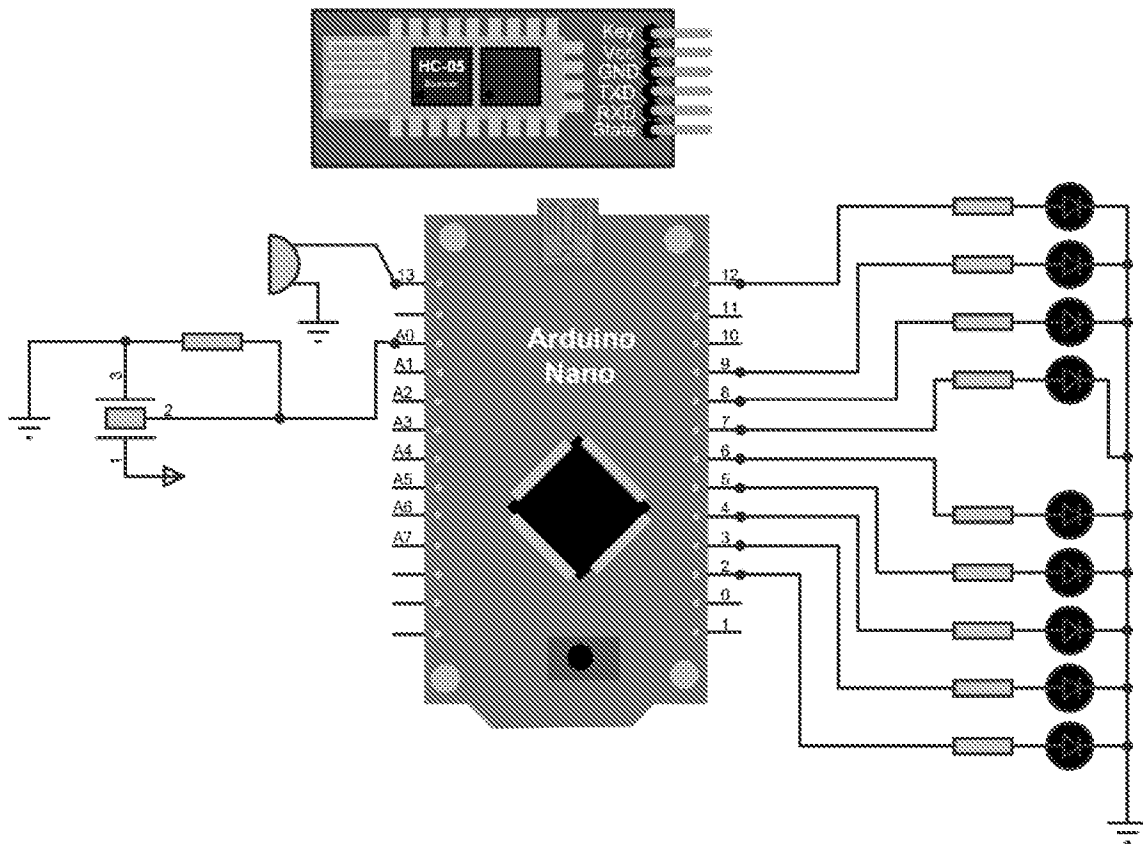
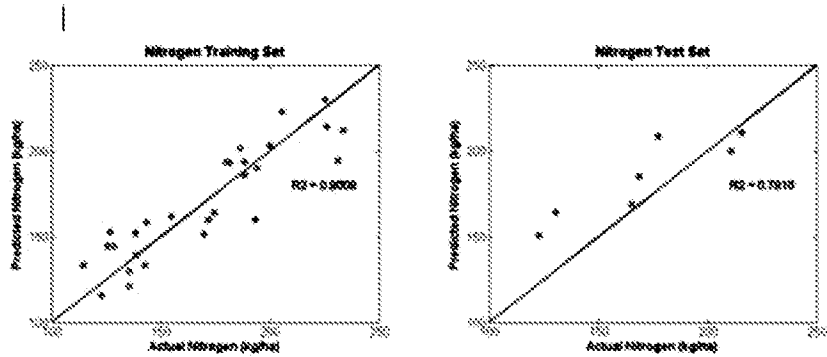


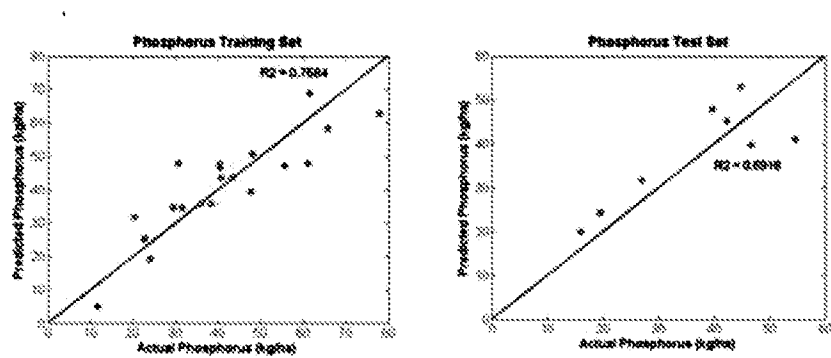
Figure 7



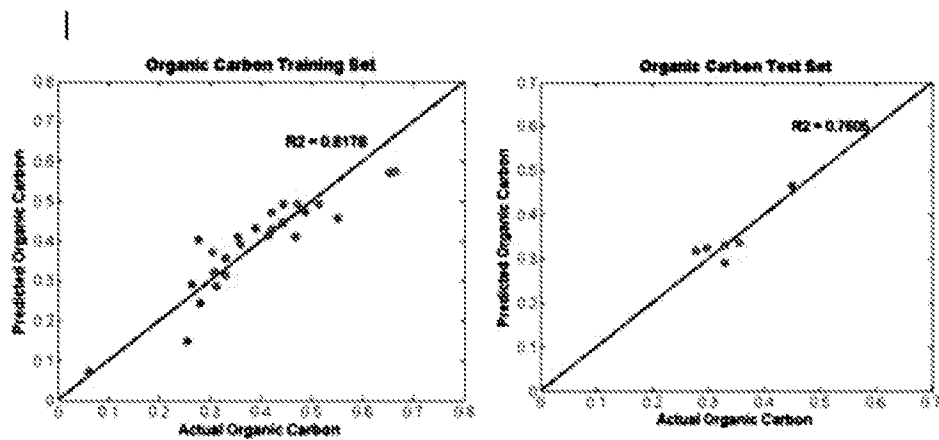
Figure 8



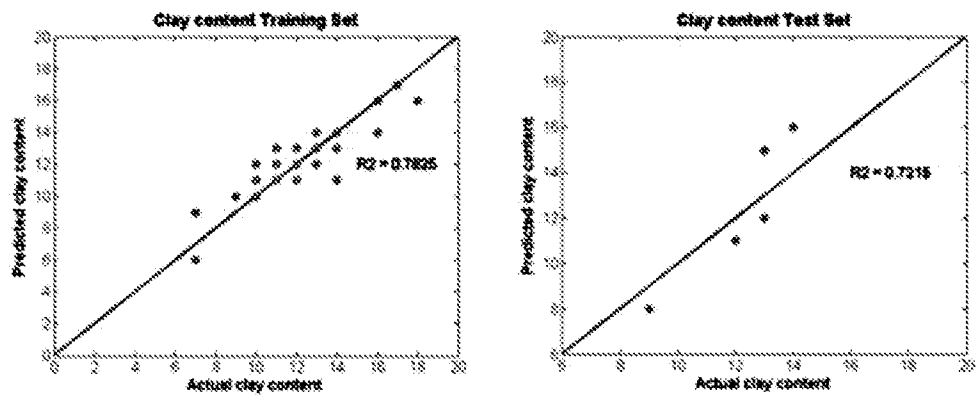
Graph 1



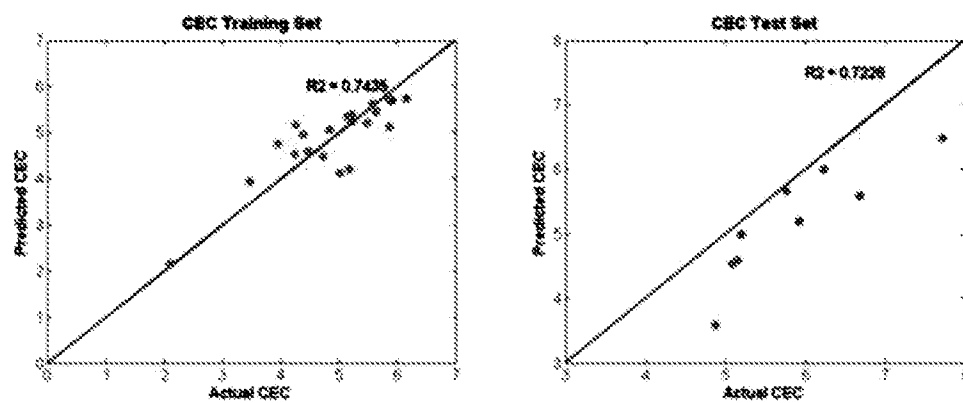
Graph 2



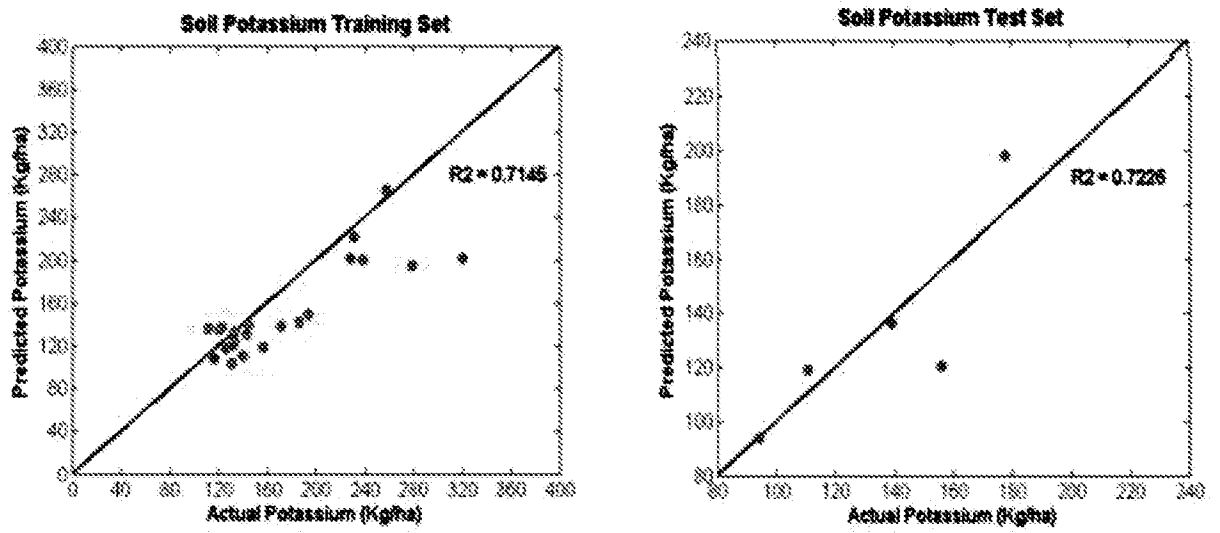
Graph 3



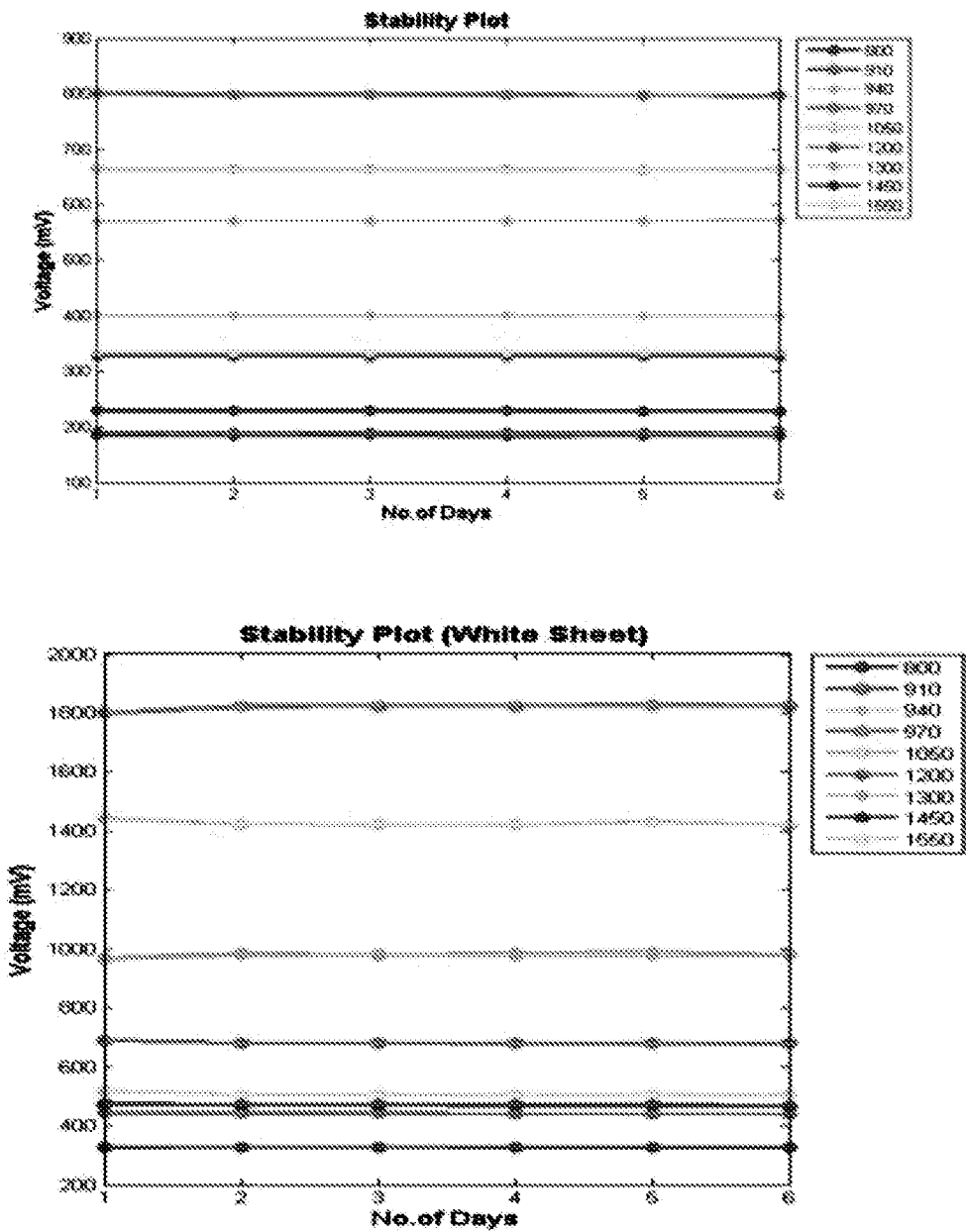
Graph 4



Graph 5



Graph 6



Graph 7

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2021/062229

A. CLASSIFICATION OF SUBJECT MATTER

G01N21/35, A01C21/00, G01N33/24 Version=2022.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01N, A01C

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

Search databases: PatSeer, IPO Internal Database.

Keywords: estimation, soil, nutrients, spectroscopy.

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	US 10107770 B2 (Ramakrishna Mission Vivekananda University, Texas Tech University System) 23 October 2018 (23.10.2018). Abstract; Column 4, Lines 17-21; Column 6, Line 58 - Column 7, Line 12; Column 21, Line 55 - Column 22, Line 7; Column 29, Lines 49-51; Figures 1, 2.	1-5
Y	US 10492361 B2 (360 Yield Center LLC) 3 December 2019 (03.12.2019). Column 6, Lines 36-65; Column 7, Lines 55-60.	1-5

☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"D" document cited by the applicant in the international application

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

31-03-2022

Date of mailing of the international search report

31-03-2022

Name and mailing address of the ISA/

Indian Patent Office
Plot No.32, Sector 14, Dwarka, New Delhi-110075
Facsimile No.

Authorized officer

Dr. Santanu Acharya

Telephone No. +91-1125300200

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/IB2021/062229

Citation	Pub.Date	Family	Pub.Date
US 10107770 B2	23-10-2018	WO 2015195988 A1	23-12-2015